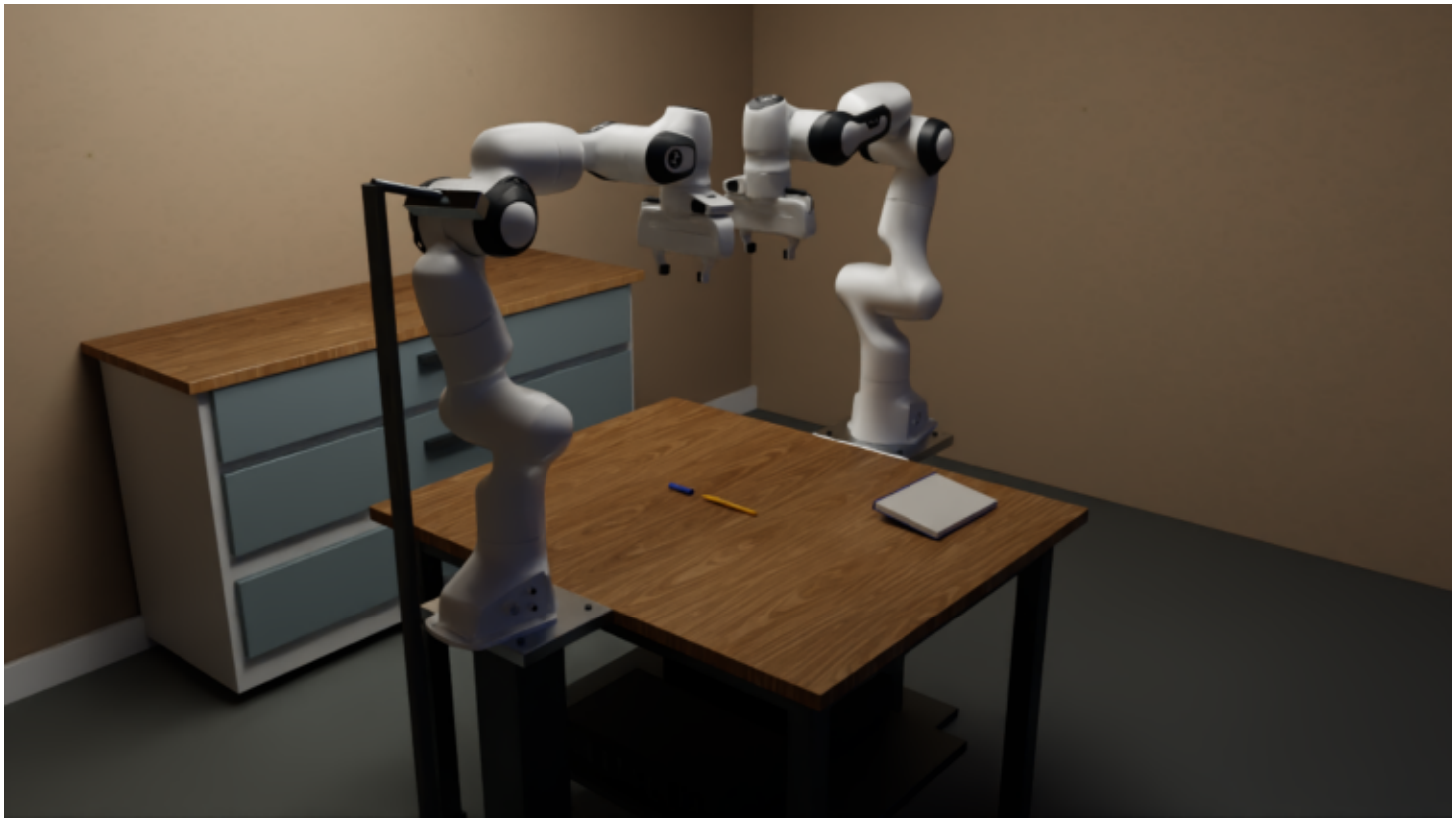
Pen-cap on Newton 1.5

Environment creation, not an autonomous task solution



Two standard Panda arms, the task's original object dimensions and positions, physical mounts, indoor PBR lighting and a represented external camera.

Full runtime	99.57 wall s / simulated s
Observed inventory	2 arms + pen + cap + notebook + room
Recording	960 Hz physics/history; 30 Hz RGB
Physical interaction	Insertion, retention, removal, grip/lift/release

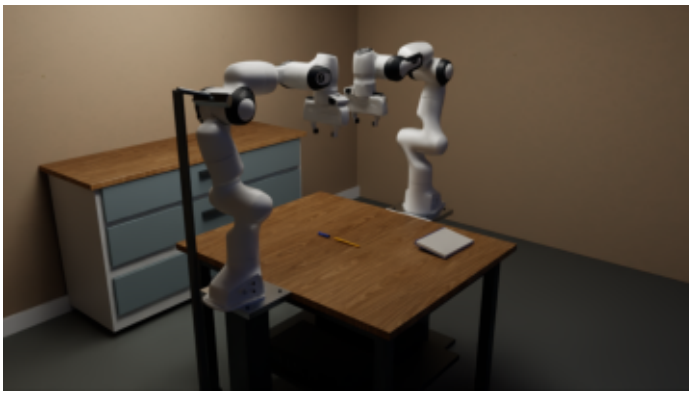
Measured removal force is 8.98 N in the 1 ms snap fixture. Videos explicitly identify their assistance.

Objects, geometry and mounting

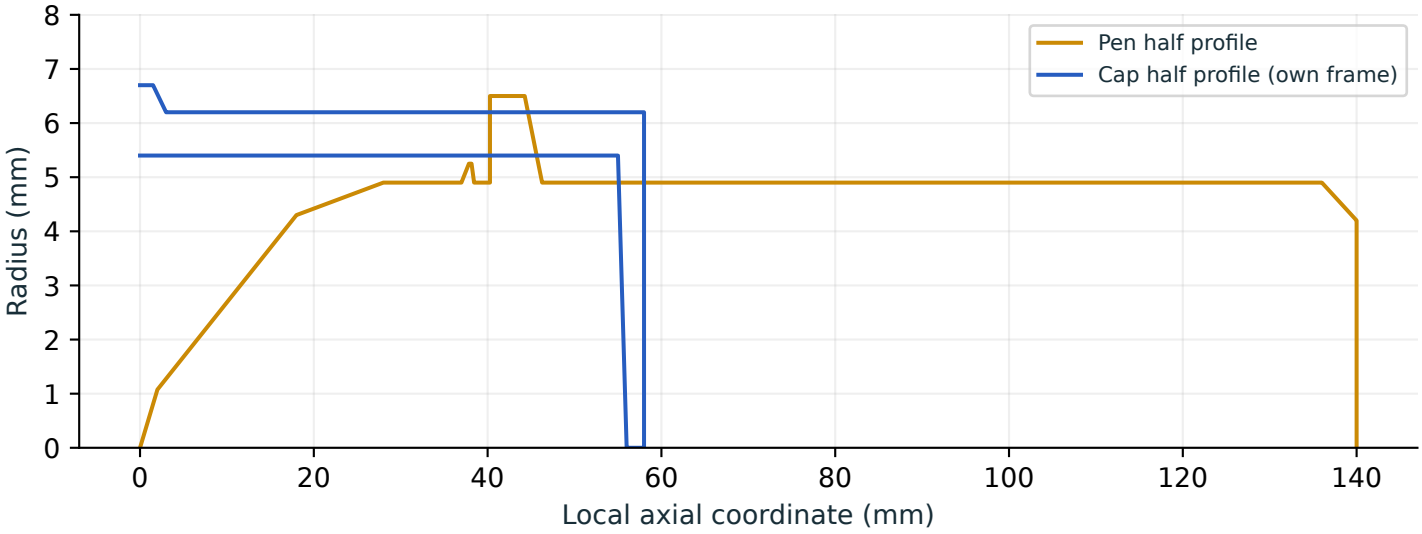
The task is not rearranged to make capping easier



Pen and cap close-up



Newton workshop scene

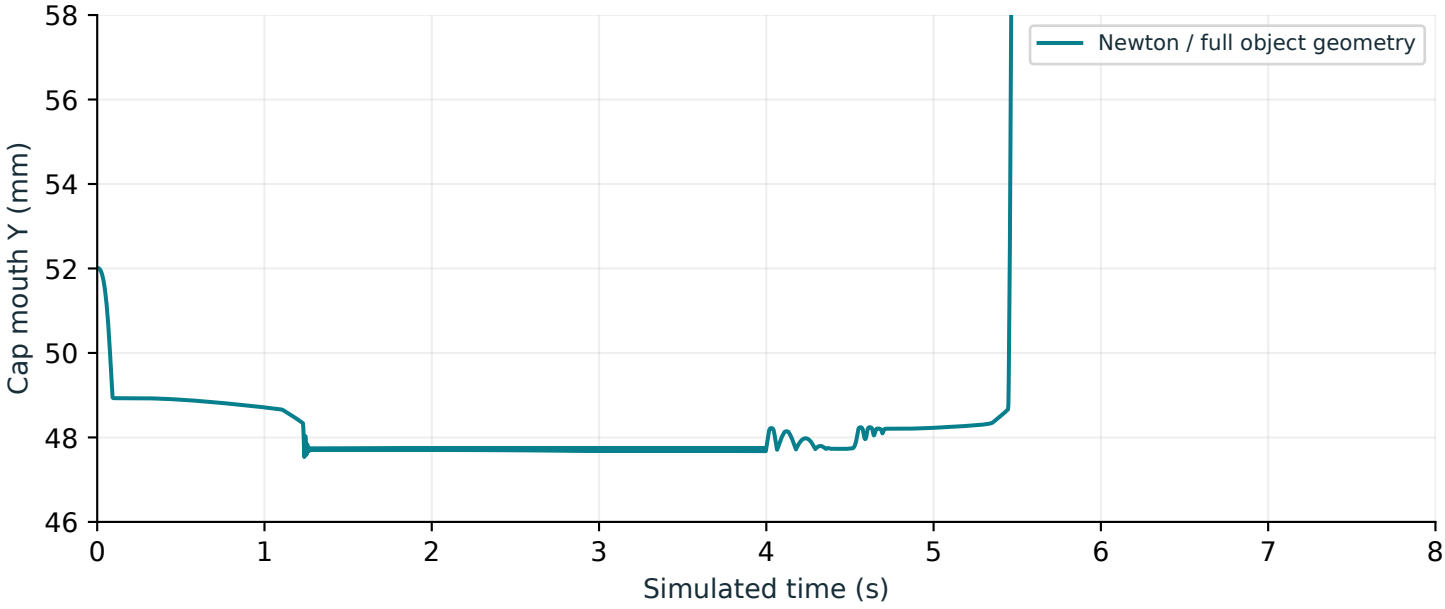


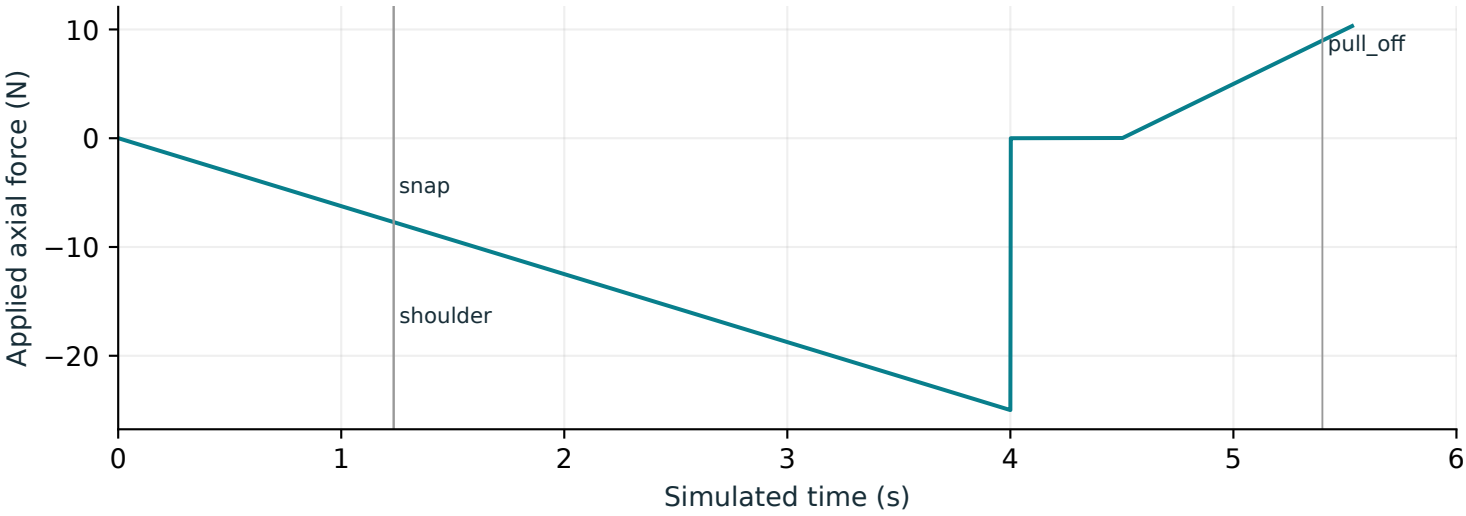
Pen: 140 mm, 5.8 g. Cap: 58 mm, 5.0 g. Bore radius 5.40 mm; bead radius 5.25 mm. Four 0.22 mm-radius nubs provide 0.15 mm radial interference. These are inherited authored parameters, not a measured commercial product pair.

The cap is a hollow nonconvex solid. Lip elasticity is approximated by contact springs (30 kN/m, 600 Ns/m per nub), not a deforming shell. No weld or pose attraction exists in the policy environment. Furniture supports, camera mount and robot bases have physical collision geometry.

Snap mechanics: measured, not latched

Full 140 mm pen / 58 mm cap, assisted force-ramp fixture





At 1 ms: insertion 7.71 N, shoulder contact 2 ticks later, removal 8.98 N. At the actual 1/960 s runtime timestep: insertion 7.53 N, removal 8.90 N.

Assistance: fixed barrel, axial rail, 300 g welded hand surrogate; its weight is supported. Applied force is logged, not mislabeled as a measured contact force. The cap remains independent of the barrel. The safety stop occurs after release under continued pulling, at 3 m/s.

Robot contact, lift and release

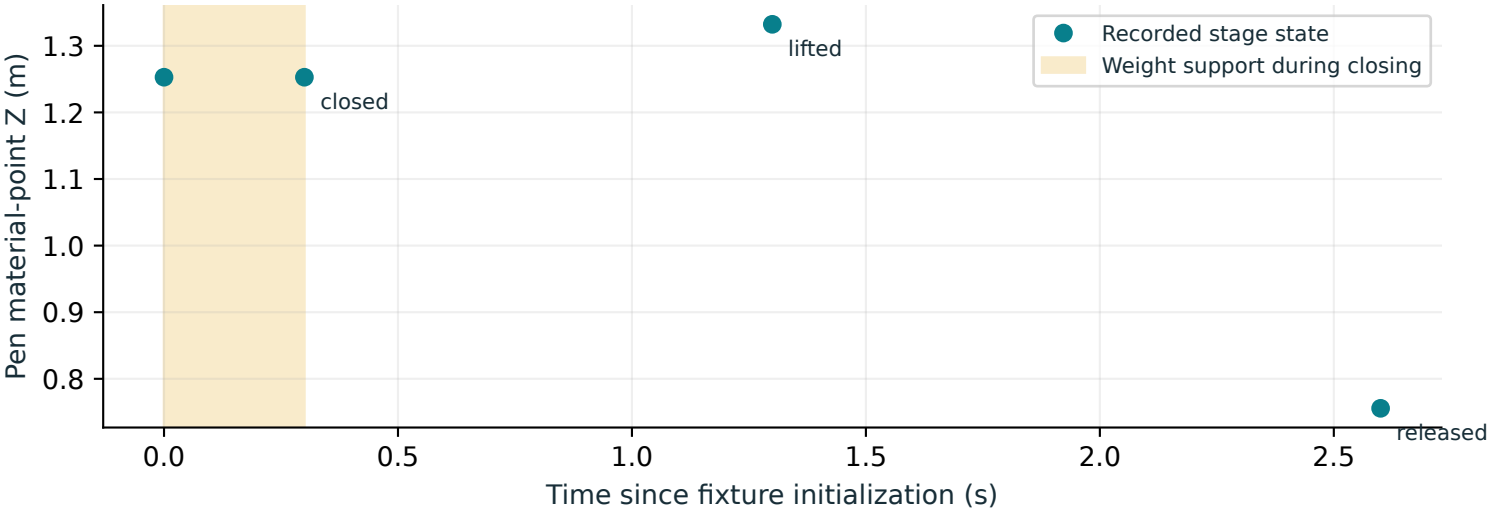
One continuous fixture after explicitly assisted initialization



After closing



After lifting: pen pivots

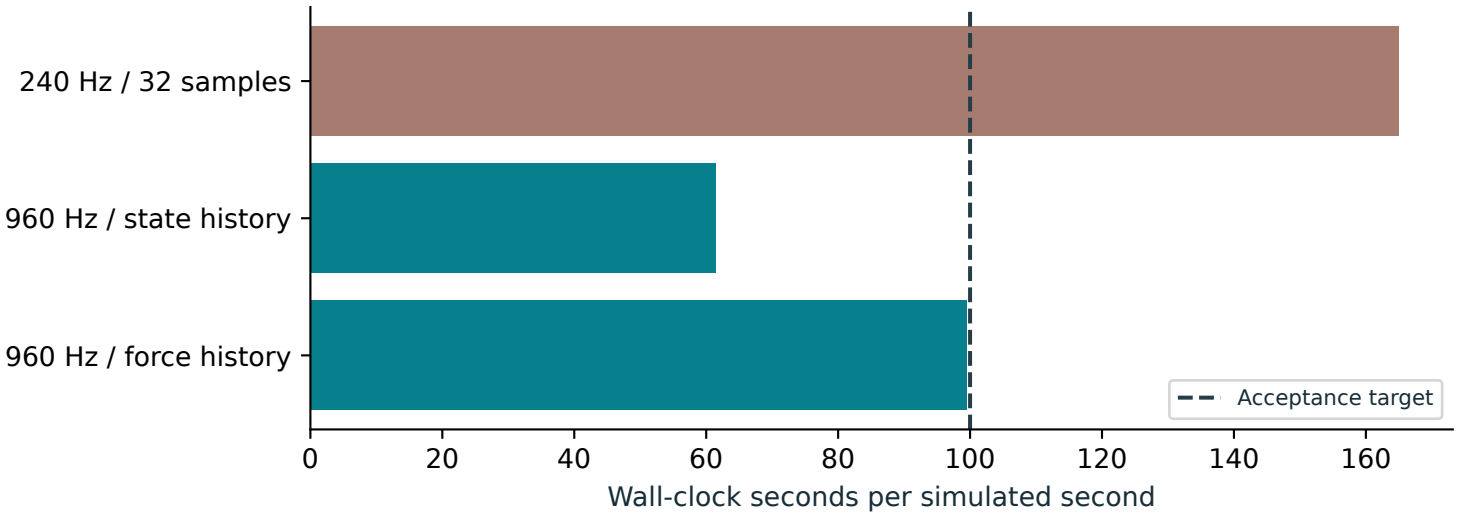


Initialization places the pen between 13 mm-open fingers, 12 mm behind the fingertip TCP. Support is removed before the robot lift. A 100 mm IK command raises the tracked pen point by 79.4 mm; the pen pivots within the grasp rather than becoming rigidly attached. Opening releases it onto the table.

The translated gripper controller preserves the 8 N effort limit and reaches 40.000 mm per-finger opening. This evidence establishes a physical grip/lift/release interaction, not an autonomous table pickup.

Performance and remote operation

Complete runtime cost, not a small-component throughput claim



Batch size 1, NVIDIA RTX 3090. The final 2-second run includes 1920 physical ticks, 61 images at 1280 x 720, and 1920 privileged history samples. Warm renderer initialization adds 11.03 s. Physics runs at 960 Hz because a 240 Hz fixture penetrates the shoulder. Object values are the same at both rates.

Verified against a real ephemeral HTTP server: authorized help/state/geometry; policy denials; duplicate action executed once; exact 24 ticks; complete per-tick trace; reset; 480-tick IK command; an actual RGB render; stale state rejected before pairing with a new image.

Access	Teacher	Restricted policy
RGB + encoders + robot actions	Yes	Yes
Object poses / mesh geometry	Yes	No
Reward / history / contact guards	Yes	No
Episode reset	Yes	No

History records states, contact witnesses, native contact forces and step impulses. Data retention is bounded and disclosed. A failed observation cannot turn an already executed action into a safe retry.

Research, limits and reproduction

No hidden calibration requirement or solver-completion gate

Model: independent rigid objects; hollow cap; interference geometry; contact-based retention; friction; standard robot geometry and mass; ordinary actuator commands.

Newton implementation: native engine/contact translation; four-pair closest-point adapter; contact material priority; hard numerical damping; stable gripper servo; 960 Hz physics; supported scene; camera/API/history integration.

Not claimed: measured product identity, deformable PP, acoustic click, air pumping, wear/fracture, rigidly locked grasps, collision-free IK paths, checkpoint/variation catalogue, or an autonomous capping solution.

The Covestro design guide supports the snap mechanism and geometry/friction dependence. Its annular formula is P = y*d*Es*X. The simulation implements a lumped contact approximation, not that continuum equation. Parameters are researched estimates or explicitly inherited task values; no physical calibration is presented as a prerequisite.

Reproduce with the release

Install the release with the Isaac Sim 5.1 renderer in a Python 3.12 environment, then start the task with the command below. Isaac renders the scene; Newton 1.5.0 performs dynamics.

```
gar run pencap --port 8080
```

Newton 1.5 release and native collision documentation

Covestro: Snap-Fit Joints for Plastics

Yoshida & Wada: Mechanics of a snap-fit

RealSense D435 product specification